STROKE SYSTEM OF CARE

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Program Design, Clinical Pathway, and Workflow

Aligned with American Heart Association / American Stroke Association (AHA/ASA) Get With The Guidelines–Stroke and American Stroke Association International Comprehensive Stroke Center Certification

Prepared for: [King Fahad Central Hospital]

Target Certification Level: [Comprehensive]

Document Owner: Stroke Program Medical Director & Stroke Coordinator

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1. Executive Summary

This document defines the hospital's Stroke System of Care: the clinical pathway, interdepartmental workflow, governance structure, and performance measurement framework required to deliver timely, evidence-based stroke care and to achieve and sustain American Heart Association / American Stroke Association (AHA/ASA) and Joint Commission stroke center certification.

The program is designed around three principles that anchor every accreditation standard in this space: (1) rapid, protocol-driven recognition and treatment of acute stroke from symptom onset through discharge; (2) a defined, multidisciplinary chain of accountability across EMS, Emergency Department, Radiology, Neurology, Pharmacy, Nursing, and Rehabilitation; and (3) continuous data-driven quality improvement through the Get With The Guidelines–Stroke (GWTG-Stroke) registry.

- Standardize the prehospital-to-discharge stroke pathway across all points of entry.
- Meet or exceed AHA/ASA Target: Stroke time-based treatment benchmarks.
- Establish the governance, staffing, and data infrastructure required for the target certification level.
- Create a durable, auditable record for accreditation survey readiness.

2. Purpose and Scope

This document applies to all patients presenting with signs or symptoms of acute stroke or transient ischemic attack (TIA), across all entry points into the organization, including Emergency Medical Services (EMS) transport, walk-in presentation, inter-facility transfer, and in-hospital (inpatient) stroke onset. It covers the full continuum: prehospital recognition, emergency evaluation, acute treatment (intravenous thrombolysis and endovascular thrombectomy where applicable), inpatient management, secondary prevention, rehabilitation planning, and discharge transition.

The document is intended for use by the Stroke Program Medical Director, Stroke Program Coordinator, Emergency Department leadership, Neurology, Neurosurgery, Neurointervention, Radiology, Laboratory, Pharmacy, Nursing, Rehabilitation Services, EMS partner agencies, and the hospital Quality and Accreditation functions.

3. Accreditation Framework

AHA/ASA stroke certification is administered in partnership with the Joint Commission (and, in some states, DNV, ACHC, or CIHQ) under four progressive certification levels. The pathway and workflow in Sections 5 and 6 are designed to satisfy the clinical elements common to all four levels, with the certification-specific elements noted where they diverge.

Certification Level	Designed For	Core Capability Beyond Prior Level	Typical Governance Requirement
Acute Stroke Ready Hospital (ASRH)	Hospitals or emergency centers with a dedicated stroke-focused program but	24/7 stroke protocol activation, CT availability, IV thrombolysis capability, and a defined transfer agreement to a higher level of care	Designated stroke medical director and coordinator; written transfer protocol

Certification Level	Designed For	Core Capability Beyond Prior Level	Typical Governance Requirement
	limited inpatient stroke capacity		
Primary Stroke Center (PSC)	Hospitals providing the critical elements for long-term improved outcomes with inpatient stroke unit care	Dedicated stroke unit, 24/7 neurology availability, standardized order sets, and formal performance-measure reporting	Multidisciplinary stroke committee; program data abstractor
Thrombectomy-Capable Stroke Center (TSC)	Hospitals performing endovascular thrombectomy and providing post-procedural neurocritical care	All PSC elements plus 24/7 neurointerventional and neurocritical care capability and dedicated post-procedure monitoring beds	Neurointervention program oversight; case volume and outcome tracking
Comprehensive Stroke Center (CSC)	Hospitals providing the highest level of coordinated stroke care for the most complex patients	All TSC elements plus 24/7 neurosurgery, advanced neuroimaging, dedicated neuro-ICU, and participation in stroke research	Full multidisciplinary leadership team; research and outcomes infrastructure

Table 1. AHA/ASA and Joint Commission stroke certification levels (most to least advanced: CSC → TSC → PSC → ASRH).

Note: Certification requirements are periodically updated by the Joint Commission and AHA/ASA. Program leadership should validate all specific criteria, volume thresholds, and staffing requirements against the current edition of the Joint Commission Comprehensive Certification Manual for Disease-Specific Care before survey.

4. Program Governance and Structure

A certified stroke program requires a named, accountable leadership structure and a standing multidisciplinary committee that meets at a defined cadence to review performance data, case outcomes, and process gaps.

4.1 Core Leadership Roles

- Stroke Program Medical Director – physician champion accountable for clinical protocols, order sets, and outcome review.
- Stroke Program Coordinator – operational owner of workflow compliance, staff education, and registry data quality.
- Emergency Department Medical and Nursing Directors – accountable for door-to-treatment time performance.
- Neurology / Neurointervention / Neurosurgery Leadership – accountable for consultation availability and procedural readiness (level-dependent).
- Quality/Accreditation Liaison – accountable for survey readiness and regulatory alignment.

4.2 Stroke Committee

A multidisciplinary Stroke Committee — including EMS liaison, ED, Neurology, Radiology, Laboratory, Pharmacy, Nursing, Rehabilitation, and Quality — should convene at least quarterly to review GWTG-Stroke performance measures, case reviews, and improvement action plans, with minutes retained for survey evidence.

5. Clinical Pathway

The pathway below represents the standardized, end-to-end route a stroke patient follows from symptom onset to discharge and secondary prevention. It is designed to be MECE across the patient journey — every arrival route and treatment decision maps to a single, unambiguous next step — and is the reference model for order sets, EMR build, and staff education.

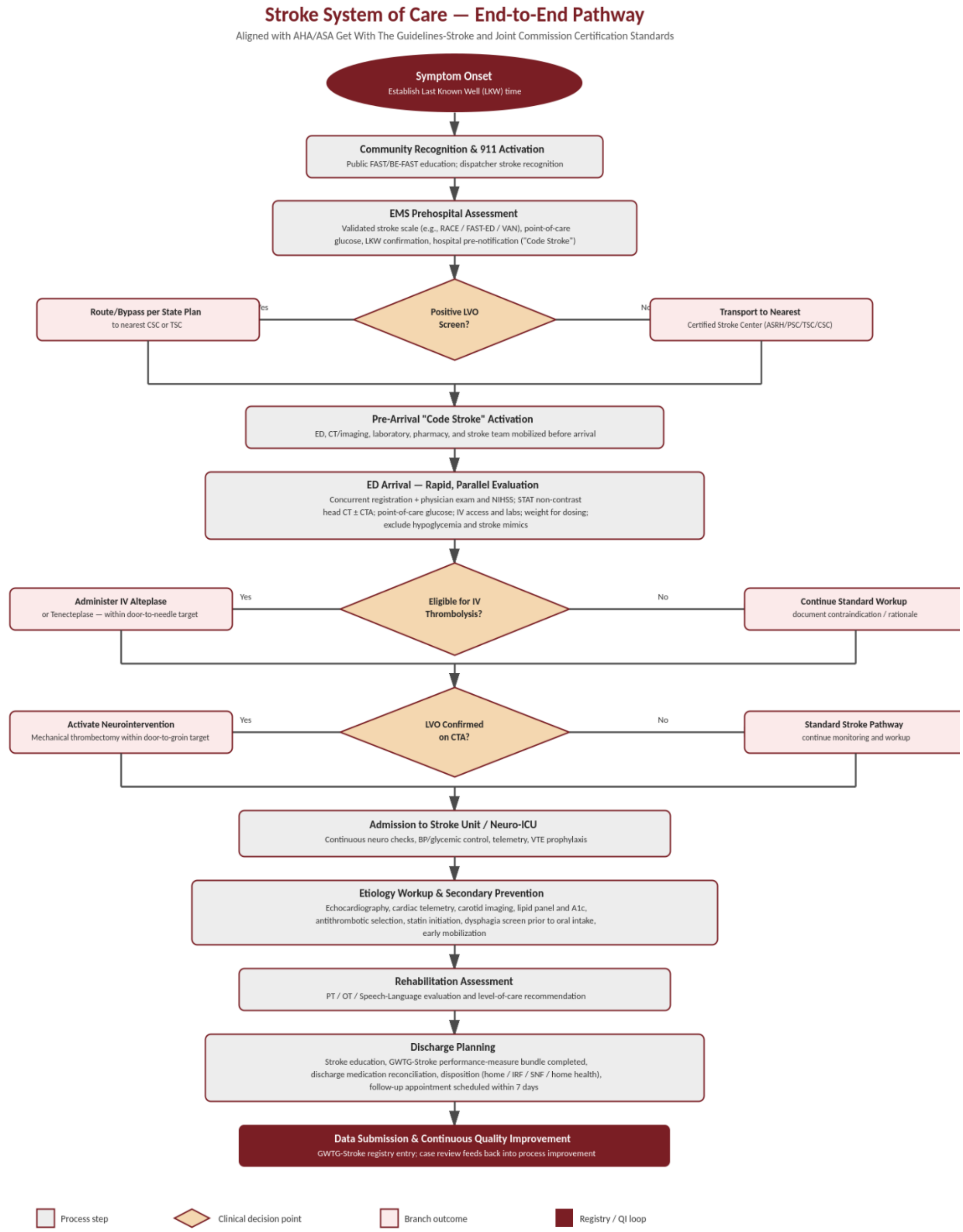

Figure 1. Stroke System of Care – end-to-end clinical pathway from symptom onset through registry submission.

5.1 Prehospital Phase

- Community education on FAST/BE-FAST symptom recognition and the importance of immediate 911 activation.
- EMS dispatch-level stroke recognition and priority response.
- On-scene validated stroke screening tool (e.g., RACE, FAST-ED, or VAN) to identify probable large-vessel occlusion (LVO).
- Last Known Well (LKW) time established and documented; point-of-care glucose obtained.
- Destination decision per state/regional EMS stroke triage and bypass protocol — direct transport to a Comprehensive or Thrombectomy-Capable center when LVO screen is positive and within transport parameters; otherwise transport to the nearest certified stroke center.
- Hospital pre-notification (“Code Stroke” alert) transmitted prior to arrival.

5.2 Emergency Department Phase

- Pre-arrival activation mobilizes the ED physician, stroke team, CT technologist, laboratory, and pharmacy before the patient arrives.
- Registration and clinical assessment occur in parallel, not sequentially.
- Standardized stroke order set initiated on arrival, including NIHSS, STAT non-contrast head CT (with CTA when LVO is suspected), point-of-care glucose, and weight-based dosing parameters.
- Eligibility for IV thrombolysis and/or endovascular thrombectomy determined per current AHA/ASA guideline criteria and documented, including rationale when treatment is withheld.

5.3 Inpatient and Secondary Prevention Phase

- Admission to a dedicated stroke unit or neuro-ICU with continuous neurological monitoring, blood pressure and glycemic control, and telemetry.
- Dysphagia screening completed before any oral intake, medication, or food.
- Etiology workup (echocardiography, carotid imaging, extended cardiac monitoring), antithrombotic and statin therapy initiated per protocol, and VTE prophylaxis.
- Early PT/OT/Speech-Language evaluation to inform level-of-care and discharge disposition.

5.4 Discharge and Transition of Care

- Structured stroke education delivered to patient and caregiver before discharge.
- Discharge medication reconciliation and completion of the full GWTG-Stroke performance-measure bundle.
- Follow-up appointment scheduled, target within 7 days of discharge.
- Case entered into the GWTG-Stroke registry to close the quality-improvement loop.

6. Emergency Department Workflow and Time-Based Targets

Time-based performance is the clinical and accreditation backbone of acute stroke care. The workflow below operationalizes the pathway into concurrent, rather than sequential, task assignments so that imaging, laboratory, pharmacy, and clinical assessment proceed simultaneously from the moment of arrival ("door").

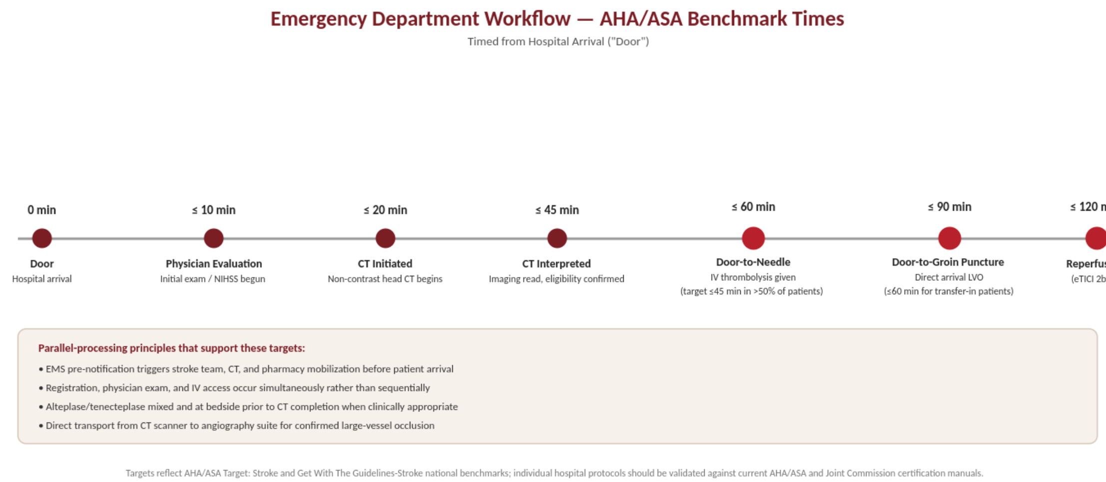

Figure 2. Emergency Department stroke workflow mapped against AHA/ASA Target: Stroke national benchmark times.

Interval	AHA/ASA Benchmark	Primary Accountable Function
Door to Physician Evaluation	≤ 10 minutes	Emergency Department triage / physician
Door to CT Initiation	≤ 20 minutes	Emergency Department / Radiology
Door to CT Interpretation	≤ 45 minutes	Radiology
Door to Needle (IV thrombolysis)	≤ 60 minutes (≤ 45 minutes in >50% of patients)	Emergency Department / Pharmacy / Neurology
Door to Groin Puncture (LVO, direct arrival)	≤ 90 minutes	Neurointervention team
Door to Groin Puncture (LVO, transfer-in)	≤ 60 minutes	Neurointervention team
Door to Reperfusion (eTICI 2b–3)	≤ 120 minutes	Neurointervention team

Table 2. Core door-to-treatment benchmark times referenced by AHA/ASA Target: Stroke and Get With The Guidelines–Stroke.

Individual thresholds and eligibility criteria should be validated against the current AHA/ASA scientific statements and the hospital's target Joint Commission certification manual, as benchmark values are updated periodically.

7. Roles and Responsibilities

Function	Key Responsibilities in the Pathway
EMS	Field stroke screen and LKW time; pre-notification; direct routing per regional bypass protocol
Emergency Department	Code Stroke activation, rapid parallel assessment, order set initiation, IV thrombolysis administration
Radiology	Immediate, prioritized imaging acquisition and interpretation for stroke alerts
Neurology / Neurointervention	Rapid consultation, treatment-eligibility determination, thrombectomy when indicated
Pharmacy	Rapid preparation and verification of thrombolytic dosing
Nursing (ED and Inpatient)	Continuous neurological monitoring, dysphagia screening, care coordination
Rehabilitation (PT/OT/Speech)	Early functional evaluation and discharge-disposition recommendation
Stroke Coordinator / Quality	Workflow compliance monitoring, registry data abstraction, case review, survey readiness

8. Performance Measurement and Continuous Quality Improvement

Performance is tracked through the GWTG-Stroke registry and reviewed at the Stroke Committee to drive an ongoing improvement cycle rather than a static, point-in-time compliance exercise.

1. Concurrent case identification and data abstraction for every stroke and TIA encounter.
2. Monthly dashboard review of door-to-treatment times and core performance measures (e.g., dysphagia screening, antithrombotic and statin therapy at discharge, VTE prophylaxis, stroke education, smoking-cessation counseling).
3. Case-level review of any patient who does not meet a time-based benchmark, with root-cause analysis.
4. Quarterly Stroke Committee review of aggregate trends and improvement action plans, with minutes retained as survey evidence.
5. Annual mock survey / readiness assessment against the current Joint Commission stroke certification manual.

9. Education and Training

- Annual stroke-specific competency validation for ED and inpatient nursing staff caring for stroke patients.
- Ongoing NIHSS certification/recertification for all clinical staff performing stroke assessments.
- Joint education sessions with EMS partner agencies on prehospital stroke scales and destination protocols.
- Community stroke-awareness education aligned with the certification level's outreach requirement.

10. Glossary of Acronyms

Acronym	Definition
AHA/ASA	American Heart Association / American Stroke Association
ASRH	Acute Stroke Ready Hospital
PSC	Primary Stroke Center
TSC	Thrombectomy-Capable Stroke Center
CSC	Comprehensive Stroke Center
GWTG-Stroke	Get With The Guidelines–Stroke (AHA/ASA quality registry)
LKW	Last Known Well (time)
LVO	Large Vessel Occlusion
NIHSS	National Institutes of Health Stroke Scale
eTICI	extended Thrombolysis In Cerebral Infarction (reperfusion grading scale)
VTE	Venous Thromboembolism
EMS	Emergency Medical Services